

CSE331 - Automata and Computability

Quiz 1

Section __ (Set A)

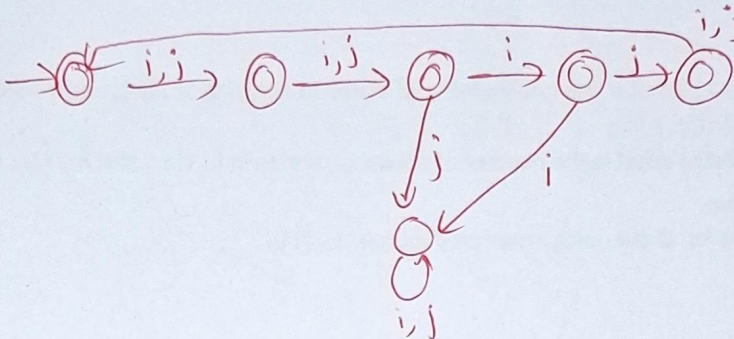
Date:

Full name:

Roll:

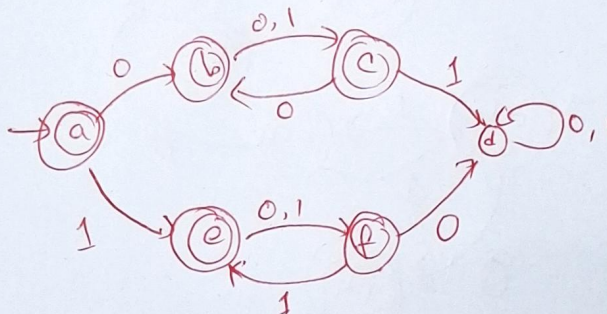
Total Marks: 20

1. (3 marks) Draw the DFA of the following language on $\Sigma = \{i, j\}$
 $L_1 = \{w \mid \text{every } 5k-2 \text{ position is } i \text{ and } 5k-1 \text{ position is } j, k > 0\}$

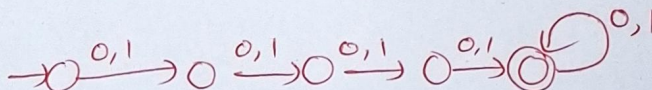


2. Draw the DFA of the following language on $\Sigma = \{0, 1\}$
 $L_1 = \{\text{odd positions contain same symbol as the first one}\}$
 $L_2 = \{\text{length of string is at least 4}\}$
 $L_3 = \{\text{sum of the symbols is odd}\}$
a. (3 x 3 marks) Draw the DFA of L_1 , L_2 and L_3

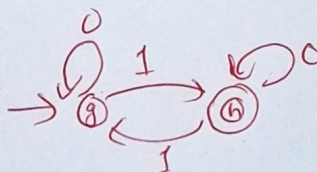
$L_1 =$



$L_2 =$



$L_3 =$



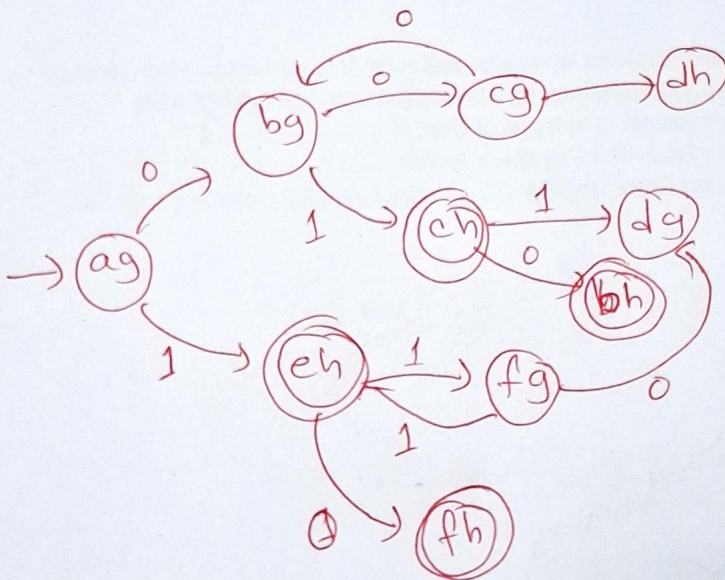
b $6 \times 2 = 12$ states

c $L_1 \cup L_2 = (5 \times 5 + 1 \times 6 - 5) = 24$ states

$\overline{L_2} \cap L_3 = (5 - 1) \text{ max} = 4$

- b. (1 mark) What is the total number of states in applying cross product rule on $L_1 \cap L_3$?
- c. (2 marks) What is the number of accepting states in $L_1 \cup L_2$ and $\overline{L_2} \cap L_3$. Briefly explain.
- d. (5 marks) Draw using cross product rule: $L_1 \cap L_3$

d)



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Quiz 1

Section __ (Set B)

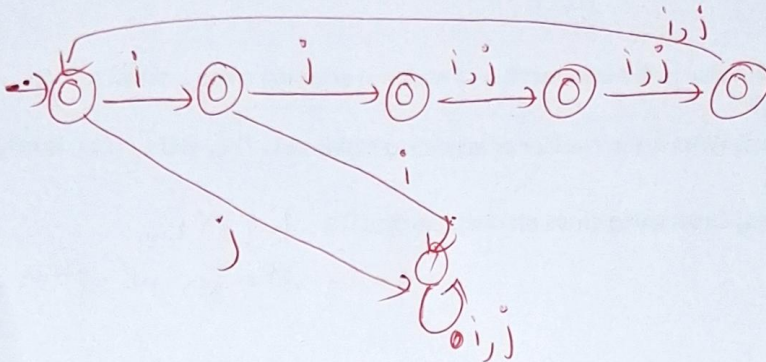
Date:

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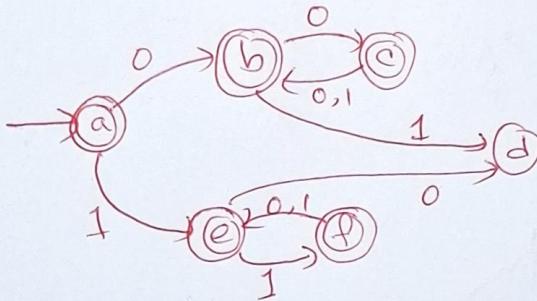
Total Marks: 20

- (3 marks) Draw the DFA of the following language on $\Sigma = \{i, j\}$
 $L_1 = \{w \mid \text{every } 5k-3 \text{ position is } j \text{ and } 5k-4 \text{ position is } i, k > 0\}$



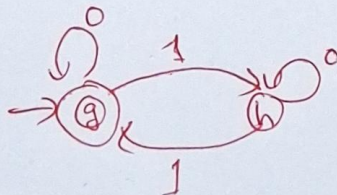
- Draw the DFA of the following language on $\Sigma = \{0, 1\}$
 $L_1 = \{\text{even positions contain same symbol as the first one}\}$
 $L_2 = \{\text{length of string is at least 4}\}$
 $L_3 = \{\text{sum of the symbols is even}\}$
 - (3 x 3 marks) Draw the DFA of L_1, L_2 and L_3

$L_1 =$



$L_2 =$

$L_3 =$



[Marks will not be deducted for using 3 states]

b $5 \times 2 = 10$ states

c $L_1 \cap L_2 = (5 \text{ max}) = 5$ states

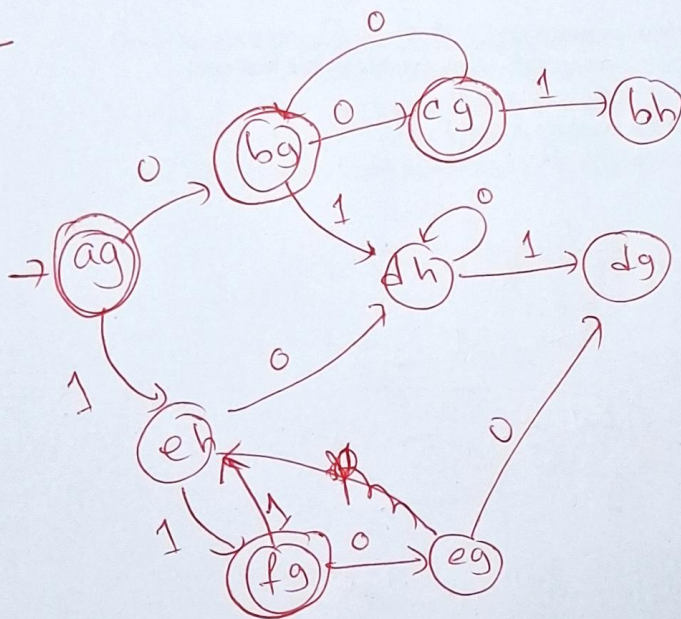
$\overline{L_2} \cup L_3 = ((5-1) \times 2 + 1 \times \overset{5}{\cancel{5}} - (5-1))$
 $= 9$ states

b. (1 mark) What is the total number of states in applying cross product rule on $L_2 \cup L_3$?

c. (2 marks) What is the number of accepting states in $L_1 \cap L_2$ and $\overline{L_2} \cup L_3$. Briefly explain.

d. (5 marks) Draw using cross product rule: ~~$L_1 \cup L_3$~~ $L_1 \cap L_3$
 (only changes accepting states)

d



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Quiz 1

Section __ (Set C)

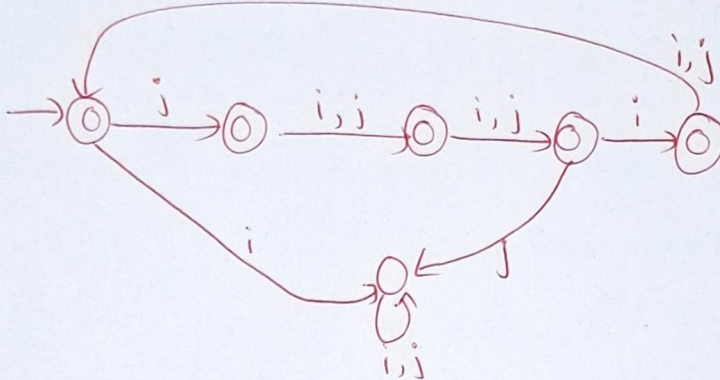
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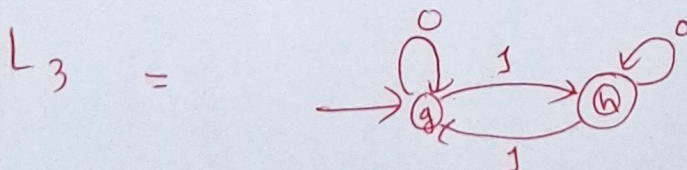
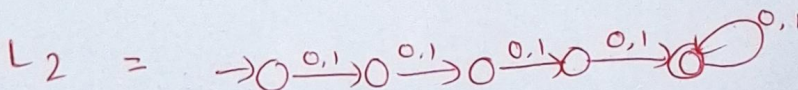
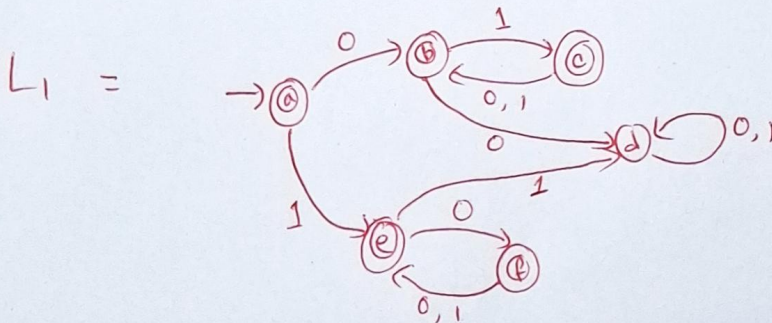
Roll:

Total Marks: 20

1. (3 marks) Draw the DFA of the following language on $\Sigma = \{i, j\}$
 $L_1 = \{w \mid \text{every } 5k-1 \text{ position is } i \text{ and } 5k-4 \text{ position is } j, k > 0\}$



2. Draw the DFA of the following language on $\Sigma = \{0, 1\}$
 $L_1 = \{\text{even positions contains different symbol from the first one}\}$
 $L_2 = \{\text{length of the string is at least 4}\}$
 $L_3 = \{\text{sum of the symbols is odd}\}$
 a. (3 x 3 marks) Draw the DFA of L_1 , L_2 and L_3



b $6 \times 2 = 12 \text{ states}$

c $L_1 \cup L_2 = (5 \times 5 + 1 \times 6 - 5) = 24$

$\overline{L}_2 \cap L_3 = (5 - 1) \text{ max} = 4$

- b. (1 mark) What is the total number of states in applying cross product rule on $L_1 \cap L_3$?
- c. (2 marks) What is the number of accepting states in $L_1 \cup L_2$ and $\overline{L}_2 \cap L_3$. Briefly explain.
- d. (5 marks) Draw using cross product rule: $L_1 \cap L_3$

